

Markscheme

Mathematics: analysis and approaches
Practice question bank

38. (a)

Consider the equation $\ln x + \ln(x + 3) = \ln 10$, where $x > 0$.

Solve the equation for x .

combine using the product law: $\ln(x(x + 3)) = \ln 10$ (M1)

$$x^2 + 3x - 10 = 0$$
A1

$$(x + 5)(x - 2) = 0 \Rightarrow x = -5 \text{ or } x = 2$$
(M1)

reject $x = -5$ since $x > 0$; $x = 2$ R1

[4 marks]

(b)

Hence find the value of $\ln x \times \ln(x + 3)$, correct to 3 significant figures.

$$\begin{aligned} & \ln(2) \times \ln(5) && \text{(M1)} \\ & = 0.693 \dots \times 1.609 \dots \approx 1.12 && \text{A1} \end{aligned}$$

[2 marks]